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FiMo-Klavier: A Multi-Stage Machine Learning System for Fine-Motor Feedback in Self-Regulated Piano Practice

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by

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To the Directors of the Program on Symbolic Systems:

I certify that I have read the thesis of Max Luis Rodriguez in its final form for submission and have found it to be satisfactory for the degree of Bachelor of Science with Honors.

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Abstract

Novice pianists must engage in deliberate, self-regulated practice to improve outside of one-on-one expert instruction. Effective practice requires the gradual development of fine-motor hand kinematics associated with stable, efficient, and technically advanced playing. This thesis proposes a multi-stage machine learning (ML) pipeline that supports such practice by generating actionable, technique-focused feedback from standard bird’s-eye-view video. The system combines a hybrid BiLSTM–Transformer anomaly detector for modeling pianist motion sequences, a hidden Markov model (HMM) layer for localizing interpretable technical evidence, and a gated large language model (LLM) feedback stage that converts selected evidence into concise corrective or affirming instruction.

Results show that a transformer-based anomaly detector can distinguish motion patterns associated with advanced play from those associated with lower-level piano performance. LLM-as-a-judge comparisons further revealed measurable semantic overlap between synthetic feedback and human-annotated reference feedback in broad technical categories such as finger curvature, wrist mechanics, and excess tension. These findings indicate that interpretable, multi-stage ML pipelines offer a promising approach for supporting self-regulated fine-motor technical development in piano practice.

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1 Introduction

The human hand is composed of 27 bones and 34 muscles, whose coordinated movement makes piano performance one of the most technically demanding forms of fine-motor skill. The acquisition of virtuosic piano skill is therefore a long and arduous process, requiring the intensive, long-term conditioning of these elementary components across both the left and right hands. Since much of this conditioning unfolds outside the lesson room, expert pedagogical oversight of independent practice is critical—without it, technical errors can quietly compound and stunt a pianist’s development [1]. For self-taught pianists, this risk is especially acute, as improper technique can develop and solidify without any corrective oversight.

Over time, improper technique can give rise to inefficient fine-motor kinematics, causing novice pianists to experience muscular fatigue more quickly and making certain passages in intermediate to advanced repertoire difficult or impossible to execute [2]. This risk is particularly pronounced in under-served communities, where access to high-quality pedagogical resources, including expert instructors, is often limited. Once students begin developing weak technical habits, long-term musical growth may be hindered: piano-pedagogy research associates inefficient technique and poorly developed practice habits with injury risk, while motor-learning research suggests that repeated practice can make movement patterns increasingly automatic and resistant to revision [3, 4, 5]. These challenges motivate the development of an accessible, on-demand artificial intelligence (AI)-based feedback tool capable of identifying technical errors early while promoting optimal kinematics that afford advanced play, thereby supporting more effective and efficient skill development.

Purely video-based computer vision pedagogical systems offer an accessible alternative for aspiring music students facing geographical or financial barriers, for whom in-person expert instruction may be prohibitively expensive or entirely unavailable [6]. To address this accessibility gap, the multi-stage pipeline proposed in this work operates on bird’s-eye-view (BEV) recordings of piano practice, requires neither specialized haptic sensors nor sensitive audio input, and focuses exclusively on observable fine-motor hand kinematics. Because it relies only on standardized camera positioning, rather than professional lighting, specialized keyboards, or additional sensing equipment, the system enables scalable, cost-efficient access to technical feedback that can generalize across

individual practice settings.

Building on this video-only design, this work presents a method for detecting technical deficiencies from isolated time-series representations of BEV piano practice and performance data, then translating those detections into actionable feedback for fine-motor development. The feedback mechanism is designed to provide low-level technical guidance that addresses not only *what* should be improved, but also *how* that improvement can be achieved physically through concrete adjustments to hand, wrist, and finger movement. Although cost-efficient piano feedback applications already exist, most automated learning tools primarily target expressiveness, sight-reading, and note accuracy rather than the physical mechanics of technique [7, 8]. Without explicit guidance on the motor mechanisms underlying improvement, novice musicians may reinforce inefficient habits, fail to diagnose technical faults, and ultimately stagnate in their development [1, 4]. The proposed system therefore complements existing pedagogical technologies by shifting the focus from musical output alone to the fine-motor technical refinement required to produce expressive, accurate playing reliably at increasingly advanced levels.

The multi-stage fine-motor feedback system introduced in this work consists of four primary components that together move from raw visual motion data to interpretable technical instruction. First, a data preprocessing and normalization pipeline extracts joint-level positional and motion features from beginner, intermediate, and advanced pianist video recordings. Second, a regression-based hybrid BiLSTM–Transformer module detects anomalous motion patterns that indicate technical and kinematic divergence across both short- and long-range temporal dependencies [9, 10]. Third, a multi-branch hidden Markov model (HMM) module localizes technical strengths and weaknesses at an interpretable fine-motor level. Finally, a gated large language model (LLM) feedback stage converts probabilistic model outputs into structured natural-language technical feedback designed to be pedagogically meaningful and actionable for novice pianists. Together, these components constitute *FiMo-Klavier*, a fully automated, video-only feedback system for fine-motor technical development in piano practice.

The effectiveness of the proposed system is evaluated through a staged experimental framework. First, internal validation experiments assess the anomaly detection module’s ability to distinguish among beginner, intermediate, and advanced levels of techni-



Figure 1: Conceptual illustration of a low-cost recording setup for capturing bird’s-eye-view piano hand kinematics for FiMo-Klavier feedback generation. The setup requires only a recording device and a flexible stand.

cal performance. Second, external holdout evaluations assess the relevance and coherence of generated feedback, including structured comparisons against human-annotated reference feedback. Finally, targeted ablation studies examine the contribution of each pipeline stage to overall system performance. Together, these quantitative and qualitative analyses demonstrate the promise of fully automated machine learning-based feedback pipelines for supporting personalized fine-motor technical development during self-regulated piano practice. Although such a system cannot replace one-on-one instruction from a human expert, it can help identify compounding technical errors before they solidify into habit and reinforce what a novice pianist is doing well through precise, affirmative technical feedback.

2 Related Work

Piano instructional tools. Historically, effective pedagogical music technologies have addressed a combination of physical, cognitive, and affective limitations faced by their target user populations [11]. Recent intelligent systems have become increasingly effective at supporting personalized learning needs in cognitive and affective domains; however, substantial challenges remain when these systems attempt to address individualized physical constraints, particularly at the fine-motor scale required for piano

technique development [12]. Commercial piano learning platforms, including Simply Piano and related app-based tutors, partially address this pedagogical gap by providing feedback on note accuracy, rhythm, and basic performance correctness. Yet these tools typically evaluate whether the correct musical output has been produced, rather than explaining how the physical movements required to produce that output should be refined [13, 8].

This distinction is pedagogically significant. Two students may achieve comparable note-level accuracy while relying on substantially different movement velocity patterns, wrist mechanics, and finger coordination strategies [14], in part because piano performance permits flexible organization of multi-joint movements through kinematic redundancy [2]. As a result, outcome-based feedback alone may fail to reveal whether a student is producing correct notes through efficient, sustainable technique or through compensatory movement habits that may later limit speed, control, endurance, or injury resilience. The present work therefore introduces a technique-centered feedback framework that explicitly targets fine-motor execution, providing personalized guidance on how small, meaningful adjustments to hand, wrist, and finger movement can support more energy-efficient, less strenuous, and technically reliable piano performance.

Hybrid neural sequence models for anomaly detection. The first stage of automated piano feedback is conceptually straightforward: the system must be able to distinguish technically mature movement from novice or inefficient movement. Because labeled examples of specific technical faults are difficult to collect exhaustively, especially at the fine-motor level, this setting motivates anomaly-detection approaches rather than standard multiclass classification. Prior work explores a hybrid LSTM–Transformer architecture for anomaly detection in time-series domains, using the LSTM component to capture short-range temporal dependencies and subsequent Transformer layers to model longer-range sequence structure [10]. This modeling strategy is well suited to piano pedagogy, where a movement that appears inefficient across a single note transition may function adaptively across a longer phrase by reducing strain or supporting continuity [2]. However, [10] assumes a binary classification objective, which is restrictive when anomalous data are scarce, heterogeneous, or difficult to label reliably.

TranAD introduces a Transformer-based regression framework for anomaly detection in multivariate time-series data, making it especially relevant to the piano domain, where advanced-player data are more readily available than consistently labeled novice-error data [9]. Rather than requiring explicit predefined technical-error categories, this framework produces anomaly scores and applies threshold-based detection, including the use of a generalized Pareto distribution to model extreme-value tails. Building on these ideas, the present work combines a hybrid BiLSTM–Transformer architecture with a regression-based anomaly scoring head to capture both short- and long-range anomalous motion patterns in pianist practice sequences while accommodating limited anomalous labels.

Interpretable modeling for feedback generation. Sequence-level anomaly scores are useful for identifying technical divergence, but they are not by themselves sufficient for pedagogical feedback: a novice pianist must be told where the movement appears strained and how that movement might be adjusted. To bridge this gap, the present work introduces hidden Markov models (HMMs) downstream of the neural anomaly detector to convert low-level motion irregularities into interpretable, localized technical signals. Prior work has applied HMMs to pianist finger-movement modeling for tasks such as estimating piece difficulty and analyzing performance structure [15, 16]. Although HMMs have largely been su-

perseded by Transformer-based approaches for many classification and autoregressive tasks, they remain useful in this setting because their state transitions and emission distributions can be mapped onto observable movement behaviors. This interpretability is especially valuable in the absence of large-scale paired datasets linking piano video directly to verified expert feedback, where model outputs must be transparent enough to support meaningful instructional generation [17].

At the same time, classical HMMs are limited by weaker representational flexibility relative to modern neural sequence models [18]. They may also face parameter-identifiability challenges when hidden-state dependencies are weak or when emission distributions are difficult to distinguish, particularly in high-dimensional motion-modeling settings such as fine-motor piano kinematics [19]. For this reason, the present work does not rely on HMMs as the primary anomaly detector. Instead, it uses multiple specialized HMM branches to localize interpretable evidence explaining the outputs of the upstream neural model, while leaving the main burden of anomaly detection to the hybrid BiLSTM–Transformer module. In this hybrid design, HMMs serve not as competing sequence models, but as an interpretability layer that translates neural anomaly scores into pedagogically actionable feedback.

Actionable synthetic feedback for piano pedagogy. Existing research on motor-based synthetic piano feedback has explored the use of recurrent neural network (RNN) outputs to generate general practice points and suggestions for improvement from gesture and action sequences [20]. While such systems can support self-regulated practice, they face two central limitations for fine-motor pedagogy. First, raw RNN outputs do not, by themselves, provide sufficiently interpretable evidence to identify localized kinematic divergence in hand, wrist, and finger motion. Second, direct comparison against advanced-level performance data, without intermediate thresholding or interpretive modeling, may offer limited instructional direction for individual users, particularly when reference datasets sparsely cover the relevant motion sequences.

A BiLSTM architecture has been applied to detect expressive features from audio–visual piano performance sequences, delivering feedback as numerical ratings across predefined categories of musical expression [7]. Such systems provide substantive insight into expressive performance quality, but numerical ratings primarily indicate *what* aspects of perfor-

mance require improvement rather than *how* those improvements can be achieved physically. For novice pianists, this distinction is crucial: actionable feedback must not only identify a weakness, but also connect that weakness to concrete adjustments in fine-motor execution.

The present work addresses these limitations by introducing an interpretable intermediate layer that translates artificial neural anomaly signals into structured fine-motor indicators before feedback is generated. This design allows the system to move beyond global performance assessment toward localized technical explanation, producing feedback that targets the physical mechanisms underlying inefficient or unstable movement.

Taken together, prior work reveals three requirements for effective neural piano feedback systems: a robust skill-level analysis method capable of operating under limited and single-class-dominated labeled data conditions; an interpretable fine-motor modeling module that converts numerical inference outputs into human-understandable technical signals; and a feedback generation mechanism that produces not only observations about technical execution, but also actionable guidance for physical improvement.

The present thesis addresses these requirements through *three primary contributions*. (1) It introduces a hybrid BiLSTM–Transformer detector that identifies anomalous motion relative to advanced reference data, enabling skill-level analysis under limited anomalous-label conditions. (2) It develops an HMM-based interpretability layer that localizes stream-level evidence and converts model outputs into fine-motor technical indicators. (3) It implements a language-model feedback stage that translates this evidence into concise technical feedback aimed at supporting self-regulated fine-motor learning.

3 Methods

3.1 Data

The anomaly-detection and interpretability modules require large-scale video data containing exemplar advanced piano technique. This work draws on two large-scale piano-performance datasets released since 2024: PianoVAM [21] and PianoMotion10M [22]. Both provide frame-wise motion data from which hand landmarks, including finger and wrist joint positions, are either directly available or can be reliably extracted.

Because the proposed anomaly detector models technical divergence relative to advanced reference

motion, advanced-player recordings serve as the primary reference distribution for training and calibration. Beginner and intermediate clips are used primarily for lower-level comparison, fine-tuning, and evaluation rather than as exhaustively labeled fault categories. This design is appropriate because fine-grained labels for specific technical errors are difficult to collect at scale, whereas advanced motion provides a practical approximation of stable, technically mature reference behavior.

To support the goal of building a widely accessible fine-motor feedback system, all features are derived exclusively from standard three-channel RGB video frames. All clips are recorded from a bird’s-eye view (BEV), which reduces hand overlap and potential side-angle occlusion while enabling stable tracking of both hands across time [22]. In practice, a pianist can replicate this setup using a flexible overhead camera mount positioned above the keyboard; any residual variation in exact camera angle is reduced through the normalization procedures described below.

In total, the processed data corpus contains 137.16 hours of piano-practice video across 16,846 clips. Of these, 16,739 clips come from 1,966 source videos in PianoMotion10M, totaling 116.16 hours, with PianoVAM providing the remaining 106 clips, totaling approximately 21 hours. PianoMotion10M consists of expert piano performances and therefore contributes only advanced-level clips, whereas PianoVAM provides practice clips from advanced, intermediate, and beginner pianists. The full corpus composition is summarized in Table 1.

For anomaly detection, clips are segmented using a sliding-window rule with window size $W = 64$ and stride $S = 16$, allowing each window to capture short continuous segments of hand kinematics. All clips are standardized to 30 frames per second (FPS), corresponding to approximately 2.13 seconds per window and a 0.53-second offset between consecutive windows from the same clip.

Run-specific subsets. Given the computational cost of training and evaluation across more than one million raw frame windows, the full pipeline does not use all processed samples simultaneously. Instead, separate subsets of the larger corpus are maintained for three purposes: base anomaly-detector training and cross-fold validation, *internal* holdout evaluation, and fine-tuning on lower-level segments not flagged as anomalous. The internal holdout data consists of reserved PianoMotion10M and PianoVAM clip shards from pianists excluded from the training

Table 1: Full processed corpus used by the pipeline. Raw windows are computed using the standard segmentation rule with $W = 64$ and $S = 16$.

Source	Skill level	Clips	Raw windows
PianoMotion10M	advanced	16,739	874,881
PianoVAM	advanced	70	90,468
PianoVAM	intermediate	27	47,916
PianoVAM	beginner	10	3,518
Total	all levels	16,846	1,016,783

set, ensuring that quantitative evaluations of the hybrid anomaly detector test generalization to unseen performers rather than memorization of performer-specific motion patterns.

Feature representation. Figure 2 shows the MediaPipe hand landmark representation used for hand-kinematics featurization. Each hand is represented by 21 tracked joint locations, providing a consistent frame-wise representation of finger articulation, hand shape, and wrist posture.

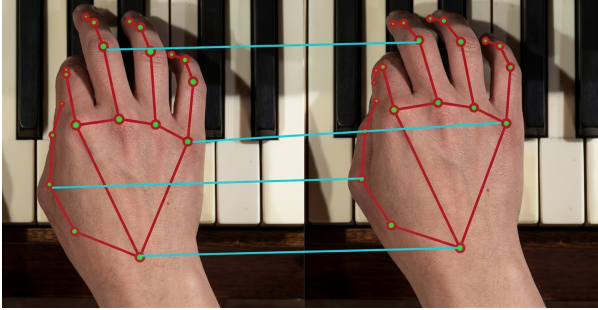


Figure 2: MediaPipe hand landmarks and inter-frame motion trajectories extracted from BEV piano practice clips.

Each processed clip stores landmarks as

$$\mathbf{P} \in \mathbb{R}^{T \times 2 \times 21 \times 3},$$

where T is the number of frames, the second axis indexes the left and right hands, the third axis indexes the 21 MediaPipe landmarks [23], and the final axis stores preprocessed normalized (x, y, z) coordinates. Let $\mathbf{p}_{t,h,j} \in \mathbb{R}^3$ denote landmark j for hand h at frame t , with $j = 0$ corresponding to the wrist. To remove absolute hand position and express each hand in a local coordinate frame, all landmark coordinates are wrist-centered:

$$\mathbf{q}_{t,h,j} = \mathbf{p}_{t,h,j} - \mathbf{p}_{t,h,0}.$$

Each frame is then compacted into a vectorized representation $\mathbf{x}_t$ by concatenating six groups of kine-

matic features. First, the wrist-centered coordinates of all non-wrist landmarks are included:

$$\{\mathbf{q}_{t,h,j} : h \in \{L, R\}, j = 1, \dots, 20\}.$$

These coordinates capture instantaneous hand shape and finger configuration. Temporal motion is represented using first- and second-order finite differences,

$$\mathbf{v}_{t,h,j} = \mathbf{q}_{t,h,j} - \mathbf{q}_{t-1,h,j}, \quad \mathbf{a}_{t,h,j} = \mathbf{v}_{t,h,j} - \mathbf{v}_{t-1,h,j},$$

for all non-wrist landmarks $j = 1, \dots, 20$. These velocity and acceleration features capture dynamic aspects of technique, including movement smoothness, abrupt transitions, and local temporal stability.

Finger bend is encoded using two joint-angle features per finger, computed as the angle between adjacent bone-segment vectors via the normalized dot product. These angles summarize proximal and distal finger curvature for each finger chain:

$$(1, 2, 3, 4), (5, 6, 7, 8), (9, 10, 11, 12), \\ (13, 14, 15, 16), (17, 18, 19, 20).$$

This representation captures whether fingers remain curved, collapsed, extended, or unstable across frames without requiring a manually specified label for each technical fault.

Because wrist-centering removes absolute wrist position, velocity, and acceleration, two wrist-posture proxy features are added per hand. The first is a wrist-height proxy, defined as the relative depth of the wrist compared with the average depth of the index, middle, and pinky MCP joints:

$$r_{t,h}^{\text{height}} = p_{t,h,0}^z - \frac{1}{3} (p_{t,h,5}^z + p_{t,h,9}^z + p_{t,h,17}^z).$$

The second is a wrist-angle proxy, computed as the standard vector angle between the wrist-middle-MCP direction and the index-pinky MCP direction. Together, these proxies retain coarse information about wrist elevation and wrist orientation while preserving the position-invariant benefits of wrist-

Table 2: Data corpus splits, sources, and window counts used across training, validation, holdout, and inference stages.

Purpose	Source composition	Clips / videos	Windows
Advanced train (base detector)	8,340 PianoMotion10M + 50 PianoVAM advanced	8,390 clips	33,560 train windows
Advanced validation	2,048 PianoMotion10M advanced	2,048 clips	8,192 calibration windows
Advanced internal holdout	6,351 PianoMotion10M + 10 PianoVAM advanced	6,361 clips	339,075 raw windows
Beginner internal holdout	PianoVAM beginner	2 clips	1,429 raw windows
Intermediate internal holdout	PianoVAM intermediate	4 clips	7,517 raw windows
Lower-level inference pool	PianoVAM lower-level train split	25 clips (6 beginner, 19 intermediate)	34,635 raw windows
Fine-tuning subset	accepted lower-level windows only	–	4,096 sampled windows
External YouTube holdout	public BEV beginner videos	20 videos	485 windows

centering.

Finally, interfinger spacing is captured using all pairwise Euclidean distances between the five fingertips:

$$\begin{aligned}\mathcal{F} &= \{4, 8, 12, 16, 20\}, \\ d_{t,h,i,j} &= \|\mathbf{q}_{t,h,i} - \mathbf{q}_{t,h,j}\|_2, \\ i &< j, \quad i, j \in \mathcal{F}.\end{aligned}$$

These distances encode finger spread and relative spacing patterns that may serve as indirect indicators of tension, collapse, or inefficient hand organization.

The total frame-level dimensionality is therefore

$$\begin{aligned}D &= 2(20)(3) + 2(20)(3) + 2(20)(3) \\ &\quad + 2(5)(2) + 2 + 2 + 2 + 2 \binom{5}{2} \\ &= 404.\end{aligned}$$

Thus, each frame is represented by a 404-dimensional vector containing non-wrist joint coordinates (120), joint velocities (120), joint accelerations (120), finger bend angles (20), wrist-height proxy features (2), wrist-angle proxy features (2), and interfinger distances (20).

Normalization. Preprocessing is applied at the clip level and kept identical across training, internal holdout, and external holdout data. This ensures that performers are compared in a common hand-centered coordinate system rather than through irrelevant variation in camera placement, global hand position, or anatomical scale. Let $\mathbf{r}_{t,h,j} \in \mathbb{R}^3$ denote the raw 3-D position of joint j for hand h at frame t , where $h \in \{\text{left}, \text{right}\}$ and $j = 0$ denotes the wrist.

The preprocessing pipeline consists of six steps.

First, missing coordinates are filled using one-dimensional interpolation applied independently to each hand, joint, and coordinate channel. Second, each coordinate channel is smoothed using a Savitzky–Golay filter with window length 7 and polynomial order 2 [24, 22]. Third, all clips are resampled to 30 frames per second. Fourth, each hand is wrist-centered:

$$\mathbf{r}_{t,h,j}^{(c)} = \mathbf{r}_{t,h,j} - \mathbf{r}_{t,h,0}.$$

This removes absolute hand position while preserving local geometry such as finger spread, curvature, and relative wrist posture.

Fifth, each hand is rotated into a canonical palm coordinate frame using a clip-level rotation matrix R_h :

$$\mathbf{r}_{t,h,j}^{(r)} = \mathbf{r}_{t,h,j}^{(c)} R_h.$$

The matrix R_h is computed from a clip-average palm basis formed from the index–pinky MCP direction and the wrist–middle-MCP direction, then orthonormalized to reduce variation caused by viewpoint and hand orientation.

Finally, coordinates are scale-normalized using a robust hand-size proxy:

$$m = \text{median}_{t,h} \left\| \mathbf{r}_{t,h,9}^{(r)} - \mathbf{r}_{t,h,0}^{(r)} \right\|_2, \quad \mathbf{r}_{t,h,j}^{(n)} = \frac{\mathbf{r}_{t,h,j}^{(r)}}{m}.$$

Here, joint 9 corresponds to the middle-finger MCP and joint 0 to the wrist. This normalization reduces sensitivity to anatomical variation across performers.

Temporal derivatives are then computed using fi-

Table 3: Augmentation configuration used for base detector training.

Augmentation	Value
Feature jitter std.	0.02
Temporal wobble probability	0.3
Maximum temporal offset	0.75 frames
Affine perturbation probability	0.3
Rotation noise std.	3.0°
Scale noise std.	0.03
Shear noise std.	0.02
Joint dropout probability	0.2
Maximum dropped joints	3
One-hand attenuation probability	0.1
One-hand attenuation range	[0.0, 0.35]

nite differences:

$$\mathbf{V}_t = \nabla_t \mathbf{r}_t^{(n)} \cdot 30, \quad \mathbf{A}_t = \nabla_t \mathbf{V}_t \cdot 30.$$

Here, $\mathbf{V}_t$ and $\mathbf{A}_t$ denote velocity and acceleration in normalized units per second and normalized units per second squared, respectively. These derivatives capture dynamic properties of technique that are not visible from static posture alone.

Feature-wise normalization. The transformer input additionally uses feature-wise normalization estimated from advanced-performance clips. For feature dimension d , let $x_{t,d}$ denote the raw feature value, μ_d the empirical mean, σ_d the empirical standard deviation, and m_d the empirical maximum absolute z-score. Each value is normalized by

$$\tilde{x}_{t,d} = \frac{x_{t,d} - \mu_d}{\max(\sigma_d, 10^{-6})}, \quad \hat{x}_{t,d} = \frac{\tilde{x}_{t,d}}{\max(m_d, 1)}.$$

The first stage performs standard z-scoring, while the second applies max-absolute scaling to prevent high-dynamic-range features from dominating training.

Augmentation for adaptation to user variability. The base hybrid anomaly detector is trained with augmentation to improve robustness to small differences in performer anatomy, recording conditions, and tracking stability. These perturbations simulate plausible geometric, temporal, and structural variation within advanced-performance motion while preserving the underlying technical pattern. All perturbations are applied before transformer feature normalization. The full augmentation configuration is summarized in Table 3.

External YouTube holdout. An *external* holdout dataset is used to evaluate pipeline feedback quality rather than the quantitative performance of individual internal modules. This dataset consists

of 20 bird’s-eye-view beginner piano-practice videos obtained from publicly available YouTube recordings. Each video is processed using the same MediaPipe landmark extraction, wrist-centering, rotation normalization, scale normalization, and feature-construction procedures applied to the training corpus. As a result, external holdout samples share the same processed representation as the training data.

During inference-time windowing, the same segmentation rule $(W, S) = (64, 16)$ is used. Additionally, when stride-aligned windows do not reach the final frame of a clip, an end-aligned full window is appended to ensure that the clip tail is evaluated.

Across the 20 external holdout videos, this procedure produced 485 scored windows.

Human-annotated evaluation of holdout. Structured technical feedback was constructed for each of the 20 external beginner holdout clips used for comparison against pipeline-generated synthetic feedback. To reduce hindsight and confirmation bias, all video annotations were completed prior to synthetic feedback generation. In addition, a predefined annotation contract was followed closely during viewing and labeling in order to improve consistency across evaluations.

The annotation contract specified a response format consisting of 2–4 sentences of practical technical feedback for the left and right hands separately. Feedback focused exclusively on observable fine-motor characteristics, including hand posture, finger movement, wrist motion, finger stability, tension release, and movement coordination. For both hands, each holdout clip was additionally assigned an overall quality label indicating whether the observed technique appeared *poor* or *good*, aligning with the pipeline’s gated positive or negative feedback generation framework. Depending on the assigned quality label, feedback emphasized either the primary technical deficiencies requiring improvement or the strongest mechanically stable aspects of the performance. However, even clips receiving positive overall evaluations could still contain targeted suggestions for further technical refinement.

All videos were evaluated with audio muted so that feedback focused exclusively on observable physical technique rather than expressiveness or tone. This design follows prior work that established structured expert feedback protocols in the expressive-performance domain, adapted here specifically to isolate fine-motor technique evaluation [7].

3.2 Anomaly Detection

Figure 3 provides an overview of the full multi-stage feedback pipeline.

BiLSTM short-range dependencies. The limited availability of publicly accessible beginner practice video discourages standard skill-level classification and motivates anomaly-based modeling. This decision is grounded in evidence that advanced pianists’ kinematics are measurably more stable and coordinated than those of beginners [14], making advanced motion a practical reference distribution for anomaly-based detection. The core detection module therefore uses a hybrid BiLSTM–Transformer architecture [10], with the BiLSTM front end providing local temporal context before transformer-based sequence modeling.

For each frame vector $\mathbf{x}_t \in \mathbb{R}^{404}$, an input projection maps the feature vector into a 128-dimensional hidden space:

$$\mathbf{z}_t = W_{\text{in}}\mathbf{x}_t + \mathbf{b}_{\text{in}}, \quad W_{\text{in}} \in \mathbb{R}^{128 \times 404}.$$

The forward LSTM summarizes past context, while the backward LSTM summarizes future context within the same 64-frame analysis window:

$$\vec{\mathbf{h}}_t = \text{LSTM}_f(\mathbf{z}_{1:t}), \quad \overleftarrow{\mathbf{h}}_t = \text{LSTM}_b(\mathbf{z}_{T:t}).$$

This bidirectional conditioning is useful for piano technique analysis because piano hand motion is governed by temporally constrained covariation across joints and digits [14], making local hand configurations more interpretable in relation to both preceding and subsequent motion [25, 26]. The two directional states are concatenated and projected into the transformer embedding space:

$$\mathbf{u}_t = W_{\text{lt}} \begin{bmatrix} \vec{\mathbf{h}}_t \\ \overleftarrow{\mathbf{h}}_t \end{bmatrix} + \mathbf{b}_{\text{lt}}, \quad W_{\text{lt}} \in \mathbb{R}^{256 \times 256}.$$

Thus, each transformer token is conditioned on short-range past and future context before long-range dependencies are modeled by the transformer encoder. The base architecture hyperparameters are summarized in Table 4.

Transformer long-range dependencies. After BiLSTM conditioning, the projected sequence $\{\mathbf{u}_t\}_{t=1}^T$ is augmented with sinusoidal positional encoding and passed through a transformer encoder. This stage allows the detector to model longer-range dependencies across the full 64-frame window, complementing

Table 4: Base architecture hyperparameters used for the hybrid anomaly detector.

Hyperparameter	Value
Input feature dimension	404
LSTM hidden dimension	128
Transformer embedding dimension	256
Number of transformer layers	3
Number of attention heads	8
Feedforward hidden dimension	512

the local temporal context captured by the BiLSTM:

$$\mathbf{g}_{1:T} = \text{TransformerEncoder}(\text{PE}(\mathbf{u}_{1:T})).$$

A first reconstruction head maps the contextualized sequence back into the input feature space:

$$\mathbf{O}_1 = f_1(\mathbf{g}_{1:T}).$$

The elementwise squared reconstruction error is then used to form a framewise focus map:

$$\mathbf{F} = (\mathbf{X} - \mathbf{O}_1)^2.$$

This focus map identifies feature dimensions and frames that are poorly reconstructed by the first pass, providing an error-conditioned representation for refinement.

In the full two-pass configuration, the second reconstruction pass concatenates the normalized input with the focus map:

$$\mathbf{R}_{1:T} = [\mathbf{X}, \mathbf{F}],$$

$$\mathbf{O}_2 = f_2(\text{TransformerRefine}(\text{PE}(W_2\mathbf{R}_{1:T}))).$$

The resulting second-pass reconstruction allows the model to refine its estimate using both the original motion features and the first-pass error structure. The one-pass ablation removes this refinement branch by setting $\mathbf{O}_2 = \mathbf{O}_1$, while the no-BiLSTM ablation removes BiLSTM conditioning and projects $\mathbf{X}$ directly into transformer space.

Training objective, scoring, and thresholding. The detector is trained to reconstruct normalized motion windows using both reconstruction heads. The training objective is

$$\mathcal{L}_{\text{train}} = \text{MSE}(\mathbf{O}_1, \mathbf{X}) + \lambda_2 \text{MSE}(\mathbf{O}_2, \mathbf{X}), \quad \lambda_2 = 1.5,$$

where λ_2 controls the contribution of the second-pass reconstruction.

At inference time, anomaly scores are computed from two-pass reconstruction error, following the

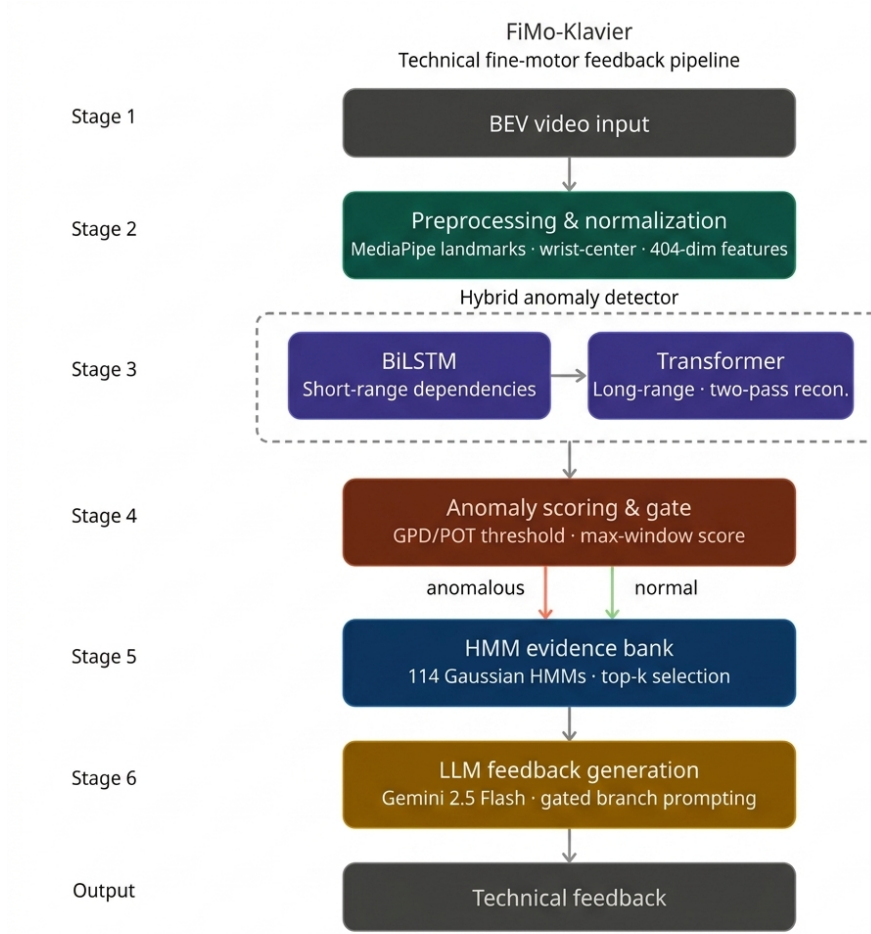


Figure 3: Overview of the multi-stage fine-motor feedback pipeline. A BEV practice video is preprocessed into 404-dimensional hand-landmark feature windows, which are passed through a hybrid BiLSTM–Transformer anomaly detector. A GPD/POT threshold gates each clip as anomalous or non-anomalous, selecting worst or best streams from a 114-model Gaussian HMM bank. A gated Gemini 2.5 Flash prompt converts the resulting stream evidence into 2–4 sentences of actionable technical feedback.

reconstruction-based scoring approach used in TranAD-style anomaly detection [9]. For a window $\mathbf{X}$,

$$s^{(1)}(\mathbf{X}) = \|\text{vec}(\mathbf{O}_1 - \mathbf{X})\|_2,$$

$$s^{(2)}(\mathbf{X}) = \|\text{vec}(\mathbf{O}_2 - \mathbf{X})\|_2.$$

The final window-level score is the average of the two reconstruction errors:

$$s(\mathbf{X}) = \frac{1}{2}s^{(1)}(\mathbf{X}) + \frac{1}{2}s^{(2)}(\mathbf{X}).$$

For a clip containing windows $\{\mathbf{X}^{(w)}\}$, the clip-level anomaly score is the maximum window score:

$$S_{\text{clip}} = \max_w s(\mathbf{X}^{(w)}).$$

A clip is classified as anomalous when

$$S_{\text{clip}} > \tau.$$

This max-window rule preserves sensitivity to short localized technical deviations and aligns clip-level inference with the window-level calibration procedure.

Threshold estimation using POT/GPD. The anomaly threshold τ is estimated using a Peak-Over-Threshold (POT) method based on the generalized Pareto distribution (GPD) [9, 27, 28]. Let $\{s_i\}_{i=1}^N$ denote calibration window scores. The tail threshold is set to the empirical 98th percentile,

$$u = Q_{0.98}(s),$$

and exceedances are defined as

$$e_i = s_i - u \quad \text{for } s_i > u.$$

If at least 10 exceedances are available and the fit is numerically stable, the exceedance distribution is modeled as

$$e \sim \text{GPD}(c, \beta).$$

The empirical tail mass is

$$p_u = \frac{|\{i : s_i > u\}|}{N},$$

where $|\cdot|$ denotes set cardinality. The target GPD tail quantile and final anomaly threshold are then computed at target tail risk $\alpha = 10^{-3}$:

$$q_\alpha = F_{\text{GPD}}^{-1}\left(1 - \frac{\alpha}{p_u}; c, \beta\right),$$

$$\tau = u + q_\alpha.$$

If the GPD fit is unstable, the fallback threshold is the empirical 99.5th percentile of calibration scores.

Cross-validation and regularization. Model selection used 2-fold cross-validation with 2 search epochs per fold to identify the best hyperparameter setting, followed by a final 4-epoch training run using those selected parameters. Optimization used AdamW with gradient clipping at 1.0 and the two-stage feature normalization described above.

The hyperparameter search varied dropout over $\{0.0, 0.1\}$ and learning rate over $\{10^{-4}, 2 \times 10^{-4}\}$, with weight decay fixed at 5×10^{-4} . Fixed architecture settings included 3 transformer layers, batch size 16, and second-pass weight $\lambda_2 = 1.5$. The selected configuration used dropout 0.0, learning rate 10^{-4} , and weight decay 5×10^{-4} .

Fine-tuning with teacher similarity and trust-region control. An additional fine-tuning stage exposes the pre-trained anomaly detector to normal, non-anomalous motion patterns from beginner and intermediate practice. This stage is motivated by the possibility that lower-level pianists may practice exercises or repertoire that differ from the advanced-performance distribution, even when their motion is not technically anomalous. The goal is to adapt the detector to additional non-anomalous motion sequences while preserving the advanced reference behavior learned during pretraining.

Fine-tuning is performed on beginner and intermediate windows whose anomaly scores do not exceed the final threshold. A frozen copy of the pre-trained

detector is retained as a teacher model, and the student model is penalized for deviating from the teacher’s second-pass reconstruction:

$$\mathcal{L}_{\text{teach}} = \frac{1}{2} \text{MSE}(\mathbf{O}_2^s, \mathbf{O}_2^t).$$

The fine-tuning objective is

$$\mathcal{L}_{\text{ft}} = \mathcal{L}_{\text{train}} + \lambda_{\text{teach}} \mathcal{L}_{\text{teach}}, \quad \lambda_{\text{teach}} = 0.5.$$

To limit drift from the advanced reference model, training monitors a normalized teacher-deviation ratio:

$$r_{\text{teach}} = \frac{\overline{\mathcal{L}}_{\text{teach}}}{\max(\overline{\mathcal{L}}_t, \epsilon)}.$$

Here, $\overline{\mathcal{L}}_t$ denotes the frozen teacher’s own reconstruction loss on the same batch. Fine-tuning is stopped early when

$$r_{\text{teach}} > 0.25,$$

indicating that the student has moved too far from the pretrained advanced baseline. This constraint reduces the risk of catastrophic forgetting of previously learned advanced motion patterns, particularly in compact transformer models with limited parameter capacity [29].

The fine-tuning stage used weight decay 5×10^{-4} and learning rate 5×10^{-5} .

3.3 Focused localization of weak and strong motion patterns using hidden Markov models

Figure 4 illustrates the HMM-based localization stage that converts anomaly detections into interpretable feedback evidence. For interpretability, the HMM bank operates on separate feature streams rather than the full concatenated motion tensor. The bank comprises 114 Gaussian HMMs spanning joint-trajectory streams, wrist proxy features, interfinger distance streams, and finger-bend streams. Each stream corresponds to a narrow, human-interpretable motion pattern, such as wrist height, wrist angle, fingertip trajectory, interfinger spacing, or finger-bend shape for a specific hand or finger.

Modeling streams independently allows each HMM to learn a compact distribution over one localized aspect of technique. This makes it possible to identify which movement pattern is least typical when a clip is flagged as anomalous, while preserving pedagogical interpretability for the downstream feedback stage. Instead of referring to an opaque anomaly score over a high-dimensional sequence, the generated feedback

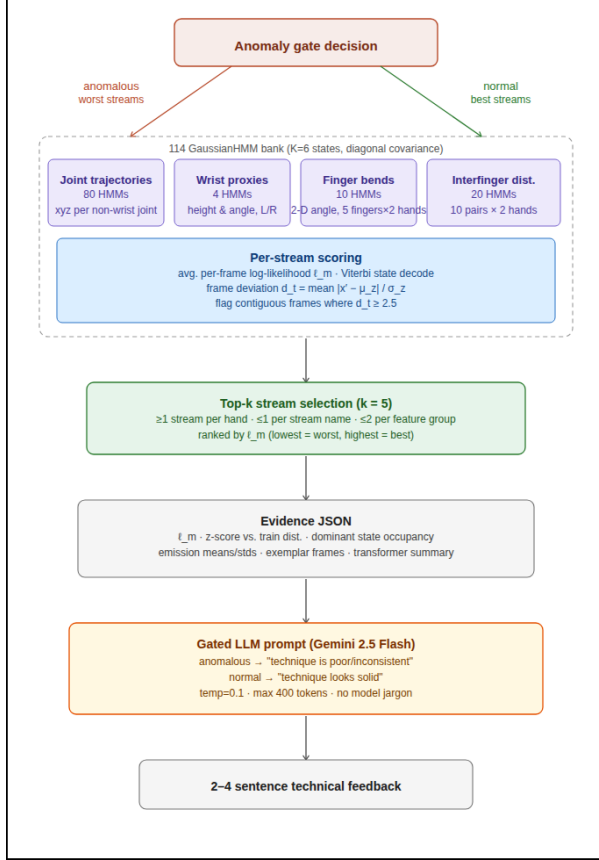


Figure 4: HMM-to-feedback stage. The anomaly gate routes each clip to weak- or strong-stream selection. Each Gaussian HMM scores one feature stream using per-frame log-likelihood and Viterbi-decoded deviation. Selected streams are converted into evidence for gated LLM feedback generation.

can refer to concrete technical categories such as unstable wrist level or inconsistent finger curvature.

Each feature stream m is normalized independently by subtracting its training-set mean and dividing by its training-set standard deviation. If $\mathbf{x}_{1:T}$ denotes the raw stream and $(\boldsymbol{\mu}_m, \boldsymbol{\sigma}_m)$ denote its scaling parameters, then

$$\mathbf{x}'_t = \frac{\mathbf{x}_t - \boldsymbol{\mu}_m}{\boldsymbol{\sigma}_m}.$$

One diagonal-covariance Gaussian HMM is trained per stream:

$$\mathbf{x}'_{1:T} \sim \text{GaussianHMM}(K = 6).$$

Each model is fit using 10 iterations of Baum–Welch expectation-maximization [30], with a minimum usable sequence length of 20 frames. The HMM bank is trained on windows drawn from 512 advanced clips in the training corpus.

Stream scoring and deviation detection. At inference time, the evidence score for stream m is the average per-frame log-likelihood:

$$\ell_m = \frac{1}{T} \log p(\mathbf{x}'_{1:T}).$$

The decoded hidden-state sequence $z_{1:T}$ is obtained using Viterbi decoding. Frame-level deviation from the decoded state is measured by

$$d_t = \frac{1}{D} \sum_{j=1}^D \left| \frac{x'_{t,j} - \mu_{z_t,j}}{\sigma_{z_t,j}} \right|,$$

where $(\mu_{z_t,j}, \sigma_{z_t,j})$ are the emission mean and standard deviation for dimension j of decoded state z_t . Frames satisfying

$$d_t \geq 2.5$$

are grouped into contiguous flagged segments. For anomalous clips, the system selects the weakest streams, defined by the lowest per-frame log-likelihoods. For non-anomalous clips, it selects the strongest streams, defined by the highest per-frame log-likelihoods.

Stream selection constraints. Stream selection follows structured diversity constraints to avoid redundant evidence. The system selects the top $k = 10$ streams while including at least one stream from each hand when available, allowing at most one instance per stream name, and allowing at most two streams from the same feature group.

3.4 Synthetic feedback generation

Synthetic feedback is generated using Gemini 2.5 Flash with a low-temperature configuration to encourage consistent, focused outputs. When needed, final expert-style regeneration uses a larger output budget to avoid truncated feedback.

The generator is prompted with compact, teacher-facing evidence from the transformer anomaly detector and HMM localization stage. The prompt frames the model as a piano-pedagogy assistant whose task is to produce concise, beginner-readable technical feedback. It preserves handedness, encourages use of evidence from both hands when available, and restricts the response to practical teaching language. It also excludes motivational filler, praise-only wording, flowery language, and references to internal model terminology such as anomaly scores, HMMs, hidden states, Gaussian emissions, log-likelihoods, z-scores, raw feature names, joint numbers, or stream labels.

Branch-specific prompting. Feedback is generated with a two-branch prompting procedure conditioned on the transformer anomaly decision. If a clip is classified as anomalous, only anomalous transformer windows are passed to the HMM bank; HMM streams are classified by lowest likelihood and the selected streams are treated as evidence of technical faults. The prompt then asks for concise corrective feedback using labels such as *Left hand (overall poor): ...* or *Right hand (overall poor):* When the evidence is mixed, the model may use *overall inconsistent*. If a clip is not classified as anomalous, all clip windows are analyzed; streams are ranked by highest likelihood, and the selected streams are treated as evidence of stable technique. The prompt then asks for affirming technical feedback using labels such as *Left hand (overall good): ...* or *Right hand (overall good): ...,* with a concise refinement when supported by evidence.

In both branches, the prompt limits feedback to 2–4 sentences, uses direct teacher-style language suitable for beginner pianists, preserves hand labels, and avoids forcing parallel feedback for both hands. Evidence time ranges are used only as internal context and are omitted from the final text unless pedagogically necessary.

One-shot exemplar structure. The generator uses one synthetic one-shot exemplar for each branch. The anomalous exemplar models concise, hand-labeled corrective feedback with *overall poor* labels and specific mechanical suggestions. The non-anomalous exemplar models concise, hand-labeled affirming feedback with *overall good* labels and a brief refinement when appropriate. These exemplars are synthetic and contain no clip-specific expert annotations, allowing them to define structure, tone, and pedagogical specificity without leaking expert labels into the generation stage.

Evidence bundle construction. The evidence bundle summarizes the information used to generate feedback. For each clip, it includes the clip-level anomaly decision, relevant transformer score statistics, the feedback branch, and the selected HMM streams after diversity filtering. Each selected stream is converted into a teacher-facing summary that identifies the hand, technical category, supporting time interval, interpreted observation or strength, pedagogical impact, and suggested correction when applicable. This structure allows the prompt to present model evidence in pedagogical terms rather than as raw scores or feature names.

This design exposes Gemini to compact pedagogical evidence rather than raw statistical outputs, allowing the generated feedback to preserve branch-specific valence, handedness, and evidence-grounded technical specificity.

4 Experiments

The following experiments evaluate the three contributions introduced in Section 2: anomaly detection under limited lower-level labels, interpretable evidence localization, and evidence-grounded feedback generation. Internal holdout analyses assess the anomaly detection module directly, while external holdout analyses evaluate the full pipeline end to end.

4.1 Ablations

To evaluate the contribution of the main anomaly-detection components, five detector variants were compared through controlled ablation experiments. Table 5 summarizes the evaluated variants.

Table 5: Detector variants evaluated in the ablation study.

Variant	Description
BiLSTM + 2-pass + aug.	Full BiLSTM front-end, two-pass transformer, with augmentation enabled.
Fine-tuned	Base detector fine-tuned on accepted lower-level windows using teacher regularization.
No augmentation	Base detector with augmentation disabled.
No BiLSTM	Transformer detector without the bidirectional LSTM front-end.
One-pass	Single-pass detector without second-pass refinement.

Beginner vs. advanced holdout classification.

The primary internal evaluation tests whether advanced clips receive lower anomaly rates than beginner and intermediate clips. The underlying assumption is that lower-level playing should exhibit more technically unstable motion than advanced performance.

Evaluation was carried out on the internal holdout split containing 339,075 advanced windows, 1,429 beginner windows, and 7,517 intermediate windows. At the window level, this setup provides a meaningful separation test for anomaly detection: advanced data establishes a large reference distribution of normal motion, while beginner and intermediate windows provide lower-level comparison data for test-

ing whether anomalous motion receives consistently higher scores and anomaly rates.

External 20-video holdout anomaly classification and synthetic feedback generation. A separate out-of-dataset evaluation was conducted on the 20 bird’s-eye-view beginner piano-practice holdout videos using all five detector variants. Each variant was used for anomaly classification, HMM evidence extraction, and synthetic technical-feedback generation on the same set of clips.

The HMM bank was shared across all variants; variant-level differences in generated feedback therefore arise primarily from each checkpoint’s anomaly scores, threshold behavior, and selected anomalous windows. For anomalous clips, only windows whose scores exceeded the saved threshold were passed to the HMM bank. For non-anomalous clips, all windows were scored and the strongest streams were used to generate affirming feedback. HMM evidence selection retained up to ten highest-ranked streams across both hands without requiring symmetric left- and right-hand output.

Generated feedback followed the hand-specific label format introduced above. The generator included only the hand sections supported by selected HMM evidence and was not required to produce output for both hands. Timestamps were available internally for evidence localization but were stripped before semantic comparison against reference annotations so that the judge evaluated technical content rather than temporal formatting.

CaSE feedback quality. To assess the quality of synthetic feedback generated during the language-model feedback stage, a CaSE-style [31] evaluation—adapted here from reasoning-chain assessment to the piano-feedback domain—was applied to feedback generated for the external 20-video holdout. CaSE parses a 2–4 sentence piece of feedback into individual steps $[\text{Step}_1, \dots, \text{Step}_N]$, and Gemini 2.5 Flash assigns binary relevance and coherence scores (0/1) to each step conditioned only on the bundle of evidence, prompt, and preceding steps; groundedness is scored on the same binary scale by the same judge, measuring how explicitly and faithfully each step is anchored in the supplied evidence. This evaluation is an internal evidence-alignment measure rather than an expert-agreement measure. It scores ordered feedback steps using the anomaly decision, selected HMM evidence, state-level divergence evidence, and sentence-level feedback segmentation.

Stepwise relevance measures whether each feed-

back step is supported by the anomaly decision and selected HMM streams. A step is marked down when feedback mentions unsupported body parts, invents timing or technical claims not grounded in evidence, or gives generic advice unrelated to selected streams. Stepwise coherence measures whether each step follows logically from previous feedback steps. Scores are reported both as stepwise aggregates and as whole-example aggregates over all 20 clips.

Because CaSE evaluates consistency with the model’s own evidence, high CaSE scores do not necessarily imply strong agreement with expert piano-pedagogy labels. This distinction is important: a response may be coherent, relevant to selected HMM streams, and well-grounded in model evidence while still missing expert concepts such as shallow keybed position, thumb tucking or flaring, inactive-finger tension, horizontal wrist rotation, or vertical wrist tilt.

4.2 Holdout evaluation against human-annotated feedback

For each model variant and each external holdout video, generated synthetic feedback was compared against combined left-hand and right-hand human-annotated reference feedback for the 20 beginner practice videos. The reference feedback was written before model comparison to reduce hindsight bias. Each reference contains separate left-hand and right-hand feedback, a per-hand overall label such as *overall poor* or *overall good*, and specific technical diagnoses, including finger curvature, wrist rigidity, thumb position, keybed depth, flared or collapsed fingers, inactive-finger tension, horizontal wrist rotation, vertical wrist tilt, fluidity, and inefficient trajectory. Timestamps were stripped from synthetic feedback before comparison, while left- and right-hand references were retained.

LLM-as-a-judge evaluation. An LLM-as-a-judge framework was used to estimate semantic overlap between synthetic technical feedback and human-annotated piano-pedagogy feedback. For each comparison, the judge received the video identifier, model variant, rubric condition, reference feedback, timestamp-stripped synthetic feedback, expert issue categories, and synthetic issue categories detected by a lightweight keyword pass. The judge returned a structured response containing a similarity score from 1 to 5, a confidence score from 1 to 5, a short rationale, and axis-wise judgments for diagnosis alignment, technical specificity, actionability, handedness accuracy, evidence faithfulness, and expert-intent alignment. The confidence score indicates certainty in the similarity

judgment, not in the anomaly detector’s decision.

To test the sensitivity of the evaluation to rubric strictness, the same fixed regenerated synthetic feedback was evaluated under three rubric settings: strict expert alignment, balanced conceptual overlap, and broad pedagogical-concept overlap. Feedback was not regenerated between rubric conditions.

Rubric conditions.

Strict expert alignment

Requires same-hand, same-valence alignment and close matching to the expert’s specific technical mechanism. Broad teacher language receives limited credit unless tied to the same concrete mechanism. Incorrect valence is penalized strongly; praising a hand labeled poor is capped at a low score unless a matching corrective mechanism is also present.

Balanced conceptual overlap

The primary middle setting. Rewards meaningful pedagogical overlap without requiring exact wording, and allows nearby concepts such as “flat fingers” and “poor curvature” to overlap. Incorrect handedness, contradictory valence, and overly generic language are still penalized. Keybed depth, thumb tucking or flaring, inactive-finger tension, wrist rotation, wrist tilt, collapsed fingers, and rigid wrists remain distinct mechanisms and do not receive overlap credit.

Broad pedagogical-concept overlap

An exploratory sensitivity setting. Credits shared pedagogical concept families—finger shape, wrist mechanics, hand position, excess tension, movement path—even when the exact expert mechanism differs. Clearly wrong hand assignment and clearly wrong positive/negative valence are still penalized; positive feedback for a hand labeled poor does not score highly unless a matching corrective concept is present.

Categorical-overlap analysis. In addition to LLM judging, a lightweight categorical-overlap analysis compared human-annotated issue categories with synthetic issue categories. This produced precision-like, recall-like, and F1-like scores over categories such as *finger curvature*, *wrist rigidity*, *horizontal wrist rotation*, *thumb position*, *shallow keybed*, *flared fingers*, *inactive-finger tension*, and *inefficient trajectory*.

This metric is intentionally approximate and is interpreted as concept coverage rather than expert agreement.

5 Results

5.1 Model selection and training

The anomaly detection module was selected using 2-fold cross-validation with 2 search epochs per fold, followed by a fixed 4-epoch final training run. The selected base configuration used a 3-layer architecture, dropout 0.0, learning rate 10^{-4} , weight decay 5×10^{-4} , batch size 16, and second-pass weight $\lambda_2 = 1.5$.

Table 6: Cross-validation and final training losses across anomaly-detector variants. Lower values indicate lower reconstruction error.

Variant	Mean CV	Best	Train	Val
BiLSTM + 2-pass + aug.	0.003006	0.001313	0.002978	0.001516
No augmentation	0.005021	0.001349	0.002810	0.001502
No BiLSTM	0.005074	0.000640	0.001480	0.000640
One-pass	0.002383	0.000656	0.001533	0.000656

Although the one-pass model achieved the lowest mean cross-validation loss, downstream anomaly-gating performance favored the augmented BiLSTM two-pass model. This indicates that reconstruction loss alone was not sufficient for model selection: the preferred detector needed to preserve separation between advanced reference motion and lower-level hold-out motion.

5.2 Threshold calibration

Table 7: Saved anomaly thresholds for each model variant.

Variant	Threshold τ
BiLSTM + 2-pass + aug.	10.6956
No augmentation	10.9788
No BiLSTM	7.5193
One-pass	12.0695
Fine-tuned	26.4784

As shown in Table 7, threshold calibration produced similar thresholds for the pre-finetuning variants, with values ranging from 7.5193 to 12.0695. The fine-tuned checkpoint produced a substantially higher threshold of 26.4784. The base model calibration used $N = 8,192$ advanced calibration scores, with 164 exceedances above $u = 5.0452$. The fine-tuned checkpoint was recalibrated using a larger mixed pool of 520,780 windows, consisting of 516,684 advanced

Table 8: Window-level anomaly ratio on the internal holdout.

Variant	Adv.	Beg.	Int.
BiLSTM + 2-pass + aug.	0.0161	0.0943	0.4900
Fine-tuned	0.000269	0.00489	0.000133
No augmentation	0.0156	0.0874	0.4819
No BiLSTM	0.0169	0.1097	0.5752
One-pass	0.0151	0.0447	0.4445

Table 9: Mean maximum sequence score on the internal holdout.

Variant	Adv.	Beg.	Int.
BiLSTM + 2-pass + aug.	6.4230	100.6379	24.7833
Fine-tuned	9.3731	101.3416	22.5804
No augmentation	6.5012	102.6397	25.0180
No BiLSTM	4.3421	78.9016	17.6527
One-pass	7.1813	105.5998	28.2635

windows and 4,096 accepted lower-level windows, producing 10,416 exceedances above $u = 12.3547$.

5.3 Internal holdout anomaly separation

Across all pre-finetuning variants, the internal holdout exhibits the expected ordering:

$$\bar{s}_{\text{advanced}} < \bar{s}_{\text{beginner/intermediate}},$$

where $\bar{s}$ denotes the mean clip-level anomaly score. This pattern appears consistently across both anomaly ratios and score magnitudes, confirming that the detector family learns a well-calibrated advanced-normal reference distribution rather than assigning elevated scores uniformly across skill levels.

Among the pre-finetuning variants, the augmented BiLSTM two-pass model provided the strongest overall anomaly-gating behavior. Removing augmentation did not improve separation and modestly reduced anomaly ratios for lower-level clips, suggesting that geometric and temporal perturbation during training improves sensitivity to lower-level motion at the cost of a marginal increase in the advanced false-positive rate, a tradeoff that favors detection of anomalous fine-motor motion. Removing the BiLSTM front end increased anomalous classifications across all levels without a compensating improvement in lower-level sensitivity, indicating that short-range temporal conditioning contributes meaningfully to the precision of the reference distribution. The one-pass ablation achieved competitive reconstruction loss but did not translate that advantage into stronger downstream gating, underscoring that reconstruction fidelity alone is an insufficient criterion for model selection in anomaly-based skill assessment.

Table 10: Trust-region history during fine-tuning.

Epoch	$\mathcal{L}_{\text{ft}}$	$\mathcal{L}_{\text{rec}}$	$\mathcal{L}_{\text{teach}}$	r_{teach}
1	0.003732	0.003569	0.000324	0.0483
2	0.002790	0.002586	0.000408	0.0607
3	0.002466	0.002238	0.000456	0.0678
4	0.002266	0.002024	0.000484	0.0721

5.4 Fine-tuning behavior

The lower-level inference stage scored 34,635 lower-level windows from 25 PianoVAM clips. Of these, 20,768 were classified as non-anomalous by the base checkpoint and 13,867 were classified as anomalous. The fine-tuning stage sampled 4,096 accepted non-anomalous windows from this pool.

Fine-tuning remained within the configured trust-region cap:

$$r_{\text{teach}} = 0.0721 < 0.25.$$

Thus, the observed behavior was not caused by optimization instability or excessive teacher-model divergence. Instead, the main change occurred at the operating point introduced by mixed threshold recalibration. The fine-tuned model sharply reduced anomalous classifications on the internal holdout, including advanced windows, but also reduced sensitivity to beginner and intermediate windows. This suggests that fine-tuning broadened the accepted motion distribution; while this may suppress false positives on simpler non-anomalous motion, it also weakens separation from the advanced reference distribution.

5.5 CaSE evaluation of synthetic feedback

Table 11 reports CaSE-style evaluation metrics for synthetic feedback generated on the external 20-video holdout. Stepwise scores average relevance, coherence, and groundedness over all judged feedback steps. Example-level scores first aggregate these values within each clip and then average across clips.

All variants scored highly on these intrinsic feedback-quality metrics, indicating that the teacher-style feedback prompt produced generally coherent and evidence-grounded feedback when conditioned on localized transformer and HMM evidence. The base augmented BiLSTM two-pass model achieved the highest stepwise and example-level relevance, while the fine-tuned model achieved the highest coherence and the no-BiLSTM model achieved the highest groundedness.

These results should be interpreted as an internal evidence-alignment measure, not as an expert-

Table 11: CaSE-style evaluation of synthetic feedback on the 20-video external holdout. Higher values indicate stronger relevance, coherence, and grounding. Bold indicates the best value in each column.

Variant	S-Rel	S-Coh	S-Grd	E-Rel	E-Coh	E-Grd
BiLSTM + 2-pass + aug	4.897	4.769	4.462	4.892	4.775	4.458
No augmentation	4.859	4.821	4.538	4.858	4.825	4.529
No BiLSTM	4.831	4.766	4.636	4.825	4.771	4.625
One-pass	4.795	4.731	4.487	4.783	4.738	4.479
Fine-tuned	4.699	4.890	4.315	4.718	4.900	4.318

agreement measure. In particular, the fine-tuned model’s high coherence reflects the internal consistency of its generated feedback, not necessarily alignment with human pedagogy judgments. This distinction becomes important in the expert-comparison results below.

5.6 External 20-video holdout classification and feedback

All five variant-specific feedback runs completed successfully on the 20 external holdout videos, with no generation failures. The resulting anomaly classification counts are shown in Table 12.

Table 12: External 20-video holdout classification counts.

Variant	Anom.	Non-anom.
BiLSTM + 2-pass + aug.	13	7
No augmentation	12	8
No BiLSTM	14	6
One-pass	11	9
Fine-tuned	0	20

All four *pre-finetuning* variants flagged a substantial and overlapping subset of the external holdout as anomalous, while the fine-tuned model flagged none of the holdout clips as anomalous. This pattern supports the interpretation that post-finetuning recalibration produced an overly permissive anomaly gate. The 11-video consensus anomaly subset across the four pre-finetuning variants consisted of videos 4, 5, 6, 7, 10, 12, 14, 17, 18, 19, and 20.

Human-annotation comparison. As described in Section 3, complete hand-specific reference feedback was available for all 20 external holdout videos. These annotations were produced by the author as an advanced pianist, rather than by independent piano-pedagogy experts, so the following results should be interpreted as an author-annotated reference comparison rather than a formal expert-panel evaluation.

For each variant and video, synthetic feedback was compared against the combined left- and right-

hand reference feedback. Timestamps were removed before LLM-as-a-judge scoring, while handedness information was retained. The judge returned a semantic similarity score from 1–5 and a separate confidence score indicating certainty in the similarity judgment, not in the anomaly detector’s decision.

Because LLM similarity scores depend on rubric strictness, three fixed judge rubrics were evaluated on the same regenerated feedback without re-generation between conditions; their definitions appear in Section 4.

Under the strict rubric, the base augmented BiLSTM two-pass model achieved the highest mean similarity (**2.15**), followed by no augmentation (2.00), no BiLSTM (1.90), one-pass (1.85), and fine-tuned (1.25). Under the balanced rubric, no BiLSTM was highest (2.25), while the base augmented and no-augmentation variants followed closely (2.20). Under the broad rubric, which credits shared pedagogical concept families even when the precise biomechanical mechanism differs, the base augmented BiLSTM two-pass model was again strongest (**2.60**, median **3.00**). Across all rubrics, the fine-tuned model remained the weakest variant, consistent with its permissive anomaly gate and resulting positive-feedback branching.

To complement the LLM-as-a-judge analysis, Table 14 reports an approximate category-overlap analysis between human annotation categories and synthetic feedback categories. For each variant, category overlap was averaged across the 20 external holdout videos and summarized using precision-like, recall-like, and F1-like scores. These scores measure coarse concept coverage rather than expert agreement.

Table 14: Lightweight category-overlap summary between human annotation categories and synthetic feedback categories, averaged across the 20-video external holdout.

Variant	Precision-like	Recall-like	F1-like
BiLSTM + 2-pass + aug.	0.53	0.53	0.51
No augmentation	0.55	0.56	0.54
No BiLSTM	0.54	0.57	0.54
One-pass	0.53	0.54	0.52
Fine-tuned	0.58	0.54	0.54

These overlap scores should be interpreted as approximate concept-coverage measures rather than direct agreement scores. They may reward broad mentions of categories such as wrist, curvature, or tension even when the generated feedback does not match the reference annotation’s hand-level valence or specific biomechanical diagnosis. For this rea-

Table 13: Sensitivity of 20-video human-annotation similarity scores to LLM judge rubric. Higher is better. Bold indicates the best mean score within each rubric.

Variant	Strict Mean	Strict Med.	Balanced Mean	Balanced Med.	Broad Mean	Broad Med.
BiLSTM + 2-pass + aug.	2.15	2.00	2.20	2.00	2.60	3.00
No augmentation	2.00	2.00	2.20	2.00	2.40	3.00
No BiLSTM	1.90	2.00	2.25	2.00	2.40	2.50
One-pass	1.85	2.00	2.10	2.00	2.35	2.50
Fine-tuned	1.25	1.00	1.55	1.50	1.65	2.00

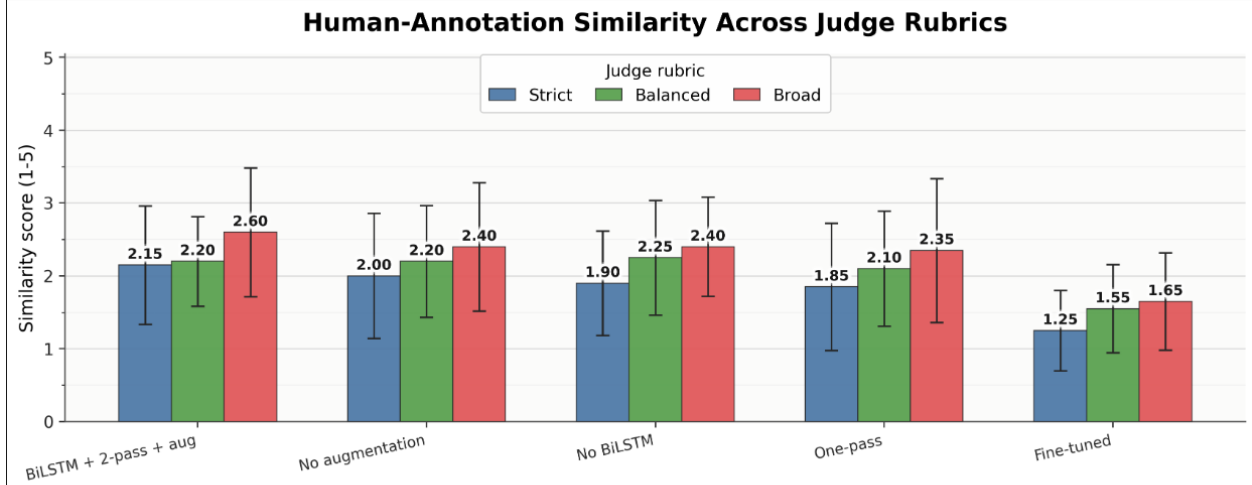


Figure 5: Sensitivity of 20-video human-annotation similarity scores to LLM judge rubric. Bars show mean similarity over 20 videos, and error bars show sample standard deviation across videos. Higher values indicate stronger semantic overlap with the human-annotated reference feedback.

son, the LLM rubric sensitivity analysis provides the stronger measure of semantic alignment.

Taken together, these results distinguish *feedback form*, *model-evidence alignment*, and *human-reference content alignment*. The strongest pre-finetuning variants recovered some coarse technical families, especially finger curvature, wrist mechanics, hand position, excess tension, and inefficient movement. However, several fine-grained reference concepts remained weakly captured, including shallow keyed position, thumb tucking or flaring, inactive-finger timing, horizontal wrist rotation, vertical wrist tilt, and collapsed knuckle structure.

6 Discussion

This discussion focuses on the two core stages of the pipeline: (1) anomaly detection and (2) feedback generation. Results for these stages are drawn from the internal and external holdout analyses, respectively.

6.1 Internal holdout analysis

Window-level anomaly ratios and threshold scores provide the primary basis for comparing detector behavior across base models and ablations. Second-pass reconstruction allows the anomaly detection

model to better reconstruct and recognize patterns in advanced playing data, lowering the anomalous classification ratio for advanced pianist playing sequence windows while imposing a stricter scoring threshold. Thus, the two-pass model uses a stricter scoring threshold while reducing anomalous classifications for advanced windows and keeping beginner and intermediate anomaly rates lower than those of the one-pass model. This suggests that the second pass helps the detector model a broader range of non-anomalous advanced motion while preserving sensitivity to extreme-value deviations.

Removing the BiLSTM increased anomalous classifications across all skill levels, suggesting that short-range temporal conditioning improves the precision of the reference distribution. Together, these results indicate that the transformer captures longer-range sequence dependencies, while the BiLSTM helps the detector recognize locally stable technical motion across shorter subsequences of playing. In practice, this allows the model to better distinguish genuinely unusual motion from technically sound clips whose broader sequence structure appears less common in the training data. This is reflected by a 3.1763 increase in the scoring threshold, while advanced, be-

ginner, and intermediate window-level anomaly ratios decrease marginally.

Augmentation also provides a meaningful improvement to the base hybrid anomaly detection two-pass reconstruction-based scoring module. Specifically, augmentation introduces additional variability into the reconstruction process; thus, while this marginally increases reconstruction error as well as the anomalous ratio for advanced pianists ($0.0156 \rightarrow 0.0161$), it allows the model to tighten the anomaly threshold and remain sensitive to greater variation in the beginner and intermediate internal holdout data. In particular, beginner anomaly ratios increase from $0.0875 \rightarrow 0.0943$ and intermediate ratios from $0.4819 \rightarrow 0.4900$.

Additional data needed for finetuning. Finetuning the base model on accepted beginner and intermediate windows caused it to classify all 20 holdout videos as non-anomalous. Expert feedback, however, indicates that many of these practice clips contain clear beginner-level technical faults, suggesting that the current finetuning procedure expands the model’s notion of “normal” too aggressively. One plausible explanation stems from the fact that anomaly decisions are governed by an extreme-value threshold: if poor beginner or intermediate technique does not exceed this threshold, it is admitted into the non-anomalous pool and then used for both finetuning and threshold recalibration. This shifts the score distribution and raises the post-finetuning threshold, so clips that would have been flagged by the base model are absorbed into the adapted model’s normal range. As a result, only the most severe deviations remain detectable, while moderate but pedagogically important technical errors are no longer classified as anomalous.

A potential remedy would be to apply a more conservative filtering rule during finetuning or to use a lower post-finetuning anomaly threshold, so that only the clearest examples of normal beginner and intermediate playing are incorporated into the adaptation set. Proper evaluation of this fix will require substantially more beginner and intermediate data, particularly data that contains well-characterized examples of both correct and incorrect technique.

6.2 External holdout analysis

CaSE results. The base BiLSTM two-pass scoring model with augmentation achieved the highest scores across both stepwise and example-level aggregate relevance. This result implies that each step of feedback is more consistently tied to the underlying

evidence, which is expected for a model that more effectively distinguishes anomalous playing patterns. The base finetuned model achieved the highest coherence scores; however, this result is not particularly meaningful in isolation, because the finetuned model did not classify any of the beginner external holdout clips as anomalous. This work also does not explicitly measure potential bias in feedback quality across anomalous and non-anomalous decision branches. For this reason, the CaSE results are best interpreted as evidence of internal prompt and evidence alignment, rather than as evidence of expert-level pedagogical accuracy. Taken together, they suggest that the synthetic feedback produced by the pipeline is internally coherent and generally relevant to the supplied motion evidence.

Interpreting similarity against human-labeled reference feedback. All reference annotations were written by the author in the capacity of an advanced pianist prior to model comparison, following a pre-defined protocol designed to limit hindsight bias. They are best understood as structured practitioner annotations rather than independently validated expert ground truth. This distinction is important for the discussion: the comparison evaluates whether the pipeline recovers pedagogically relevant concepts from the same clips, not whether it has been pedagogically validated against a panel of instructors.

The representative comparisons in Figure 6 clarify the main pattern behind the similarity scores. In high-alignment cases, the synthetic feedback often identifies the same broad technical family as the human annotation, such as finger curvature, wrist mechanics, or unstable hand shape. However, the model usually expresses these issues at a coarser biomechanical level, without reference to the keyboard-relative or anatomical landmarks that human annotations characteristically invoke. For example, the reference annotation may distinguish horizontal wrist rotation, shallow keybed position, or thumb strain, while the synthetic feedback may collapse these observations into broader descriptions such as unstable wrist level or inconsistent finger curvature.

This difference suggests that the pipeline is better at recovering coarse pedagogical categories than fine-grained technique diagnoses. That limitation is expected given the current evidence representation: the HMM streams expose localized motion evidence, but they do not yet encode all of the biomechanical concepts that piano teachers routinely use, such as inactive-finger timing, thumb tucking or flaring, vertical wrist tilt, collapsed knuckle structure, or

Figure 6: Representative high- and low-alignment comparisons for the Base BiLSTM + two-pass + augmented feedback generator. The high-alignment example shows significant agreement on broad technical families, while the low-alignment example shows failure caused by non-anomalous branching.

Case	Human annotated feedback	Synthetic feedback
High alignment Video 9 Similarity: 4/5 Branch: anomalous	Left hand (overall poor): Finger curvature is poor across all fingers. Pronounced wrist rotation produces non-fluid movement. Wrist rigidity contributes to inefficient motion overall. Right hand (overall poor): The middle through pinky fingers are collapsed. The hand sits too shallow in the keybed with little curvature. The thumb is strained and sits too shallow on the keys.	Left hand (overall poor): The middle finger looks too flat, which can reduce control and make the hand feel unsupported. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.
Low alignment Video 2 Similarity: 1/5 Branch: non-anom.	Left hand (overall poor): Finger joints are stiff, producing linear, rigid movement. The index finger remains notably flat; increased curvature is needed. The hand sits too shallow in the keybed. Right hand (overall poor): The ring and pinky fingers tighten and straighten even when not playing. The hand should be positioned deeper in the keybed. Wrist movement appears strained, and the index finger requires more curvature.	Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. The hand shape stays compact and the fingertip path is controlled, helping the motion remain relaxed and efficient.

depth of contact in the keybed. As a result, the generated feedback can be pedagogically plausible and evidence-grounded while still missing the more specific mechanism identified in the reference annotation.

The low-alignment example illustrates a different failure mode. When the anomaly gate classifies a technically weak clip as non-anomalous, the feedback branch shifts from corrective to affirming language. In these cases, even fluent and well-structured feedback can become semantically misaligned with the reference because the branch-level valence is wrong. This explains why the fine-tuned model performed poorly in semantic judging despite producing polished feedback: its permissive post-recalibration threshold caused positive-feedback branching for clips that the human reference labeled as technically problematic.

These examples also show why the strict, balanced, and broad rubric settings are best interpreted as a sensitivity analysis rather than as absolute measures of feedback quality. Strict scoring captures whether the synthetic feedback matches the same hand, valence, and technical mechanism. Broad scoring captures whether the feedback shares a larger pedagogical concept family. The gap between these settings is informative: it indicates that the current system often reaches the correct region of the pedagogical space, but does not consistently recover the specific teacher-level diagnosis.

Taken together, the similarity and dissimilarity

rationales support a cautious interpretation of the pipeline’s feedback quality. The best pre-finetuning variants can generate structured, hand-labeled feedback that partially overlaps with human-labeled technical observations, especially for broad concepts such as finger curvature, wrist mechanics, hand position, excess tension, and inefficient movement. However, the system remains limited by the granularity of its evidence representation and by the reliability of the anomaly gate. Future versions should therefore improve both the technical vocabulary encoded in the HMM evidence layer and the calibration procedure that determines whether a clip receives corrective or affirming feedback.

Interpreting the elevated intermediate anomaly ratio. It is worth noting the substantially higher anomaly ratio observed for intermediate player internal holdout data from the PianoVAM dataset: 0.4900 compared to 0.0161 for advanced and 0.0943 for beginner data. Intermediate holdout windows are derived from a relatively small PianoVAM subset, and PianoVAM reports player skill level using subjective self-reported proficiency [21].

Intermediate holdout metrics follow similar directional trends across ablation conditions, suggesting that the observed behavior is structurally consistent rather than random. However, because the intermediate holdout subset is small, its absolute anomaly ratio should be interpreted cautiously even though the directional trend is consistent across ablations. This limitation does not invalidate the comparative

results but highlights the importance of collecting more balanced and objectively labeled intermediate-level datasets in future work.

7 Conclusion

The multi-stage feedback pipeline presented in this work, FiMo-Klavier, offers a promising method for providing accessible, technique-focused feedback for novice classical pianists. The hybrid anomaly detection system demonstrated the ability to distinguish lower-level from advanced-level playing at a fine-motor scale, while the downstream HMM interpretation stage provided localized motion evidence that could be translated into concrete technical feedback. Comparisons against human-annotated reference feedback suggest that Gemini 2.5 Flash feedback, conditioned on HMM-derived evidence, can recover meaningful coarse pedagogical categories such as finger curvature, wrist mechanics, hand position, excess tension, and inefficient movement. However, strict rubric scores remained modest, indicating that the current system does not yet consistently match the fine-grained specificity of human technical annotation, especially for concepts such as shallow keybed position, thumb tucking or flaring, inactive-finger timing, horizontal wrist rotation, vertical wrist tilt, and collapsed knuckle structure.

Taken together, these results support the central thesis of this work: interpretable, multi-stage machine learning pipelines can provide a feasible foundation for self-regulated fine-motor feedback in piano practice, but their pedagogical value depends on reliable anomaly gating and sufficiently granular evidence representation. The best-performing pre-finetuning variants produced feedback that was coherent, hand-labeled, and partially aligned with human-labeled technical observations, suggesting that video-only motion evidence can support useful instructional generation. At the same time, the fine-tuned model’s overly permissive anomaly gate showed that fluent feedback alone is insufficient; the system must first route clips into the appropriate corrective or affirming branch. Thus, the main contribution of this thesis is not a finished substitute for expert instruction, but an empirically evaluated framework for connecting video-based hand-motion modeling, interpretable technical evidence, and natural-language feedback generation. Critically, because all reference annotations were produced by the author rather than independent piano-pedagogy experts, the final similarity score metrics should be interpreted as a preliminary feasibility signal; independent expert validation is

the most important next step before any stronger pedagogical claims can be made.

Future work should first strengthen the data foundation for lower-level technique modeling. With additional beginner and intermediate data, it would become possible to analyze which feature streams are most frequently classified as anomalous and to identify the fine-motor technical areas that most consistently break down during self-regulated practice. A second direction is activity-based clustering, using methods such as K-means or related sequence-clustering approaches, to account for the fact that different musical passages require distinct movement types. Without this form of stratification, the model may become biased toward movements that are common in the training repertoire and may incorrectly classify rare but valid fine-motor patterns as anomalous because they are underrepresented in the dataset.

A third direction is real user evaluation. A field study with beginner pianists is necessary to determine whether generated feedback produces measurable improvements in technical skill and whether improvements in technical efficiency translate into stronger musical performance outcomes. Future evaluation should also incorporate independent piano-pedagogy experts rather than relying only on author-generated reference annotations. Finally, the lower-level fine-tuning procedure should be revised using more conservative filtering and better-calibrated inference thresholds so that only clearly non-anomalous beginner and intermediate windows are incorporated into the adaptation set.

References

- [1] A. Suzuki and H. Mitchell, “What makes practice perfect? how tertiary piano students self-regulate play and non-play strategies for performance success,” *Psychology of Music*, vol. 50, p. 030573562110109, 05 2021.
- [2] S. Furuya and E. Altenmüller, “Flexibility of movement organization in piano performance,” *Frontiers in Human Neuroscience*, vol. 7, p. 173, July 2013.
- [3] B. S. Ilari, P. Keller, H. Damasio, and A. Habibi, “The development of musical skills of underprivileged children over the course of 1 year: A study in the context of an el sistema-inspired program,” *Frontiers in Psychology*, vol. 7, p. 62, feb 2016.
- [4] I. Tchernik, “Healthy piano technique and the prevention of professional injuries: An ex-

- ploration of the schmidt-shklovskaya-minsker method and its implementation in piano pedagogy,” master’s thesis, York University, 2017.
- [5] A. M. Haith and J. W. Krakauer, “The multiple effects of practice: skill, habit and reduced cognitive load,” *Current Opinion in Behavioral Sciences*, vol. 20, pp. 196–201, apr 2018.
 - [6] F. Diller, G. Scheuermann, and A. Wiebel, “Visual cue based corrective feedback for motor skill training in mixed reality: A survey,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 7, pp. 3121–3134, 2024.
 - [7] X. Deng, “Ai-driven emotional intelligence in piano education: Deep learning models for expressive performance coaching,” *Acta Psychologica*, vol. 263, p. 106264, 2026.
 - [8] K. Lei, “The effectiveness of special apps for online piano lessons,” *Interactive Learning Environments*, vol. 31, no. 10, pp. 7455–7466, 2023.
 - [9] S. Tuli, G. Casale, and N. R. Jennings, “Tranad: Deep transformer networks for anomaly detection in multivariate time series data,” 2022.
 - [10] M. Wei, Z. Wei, W. Chen, B. Jing, and Y. Qi, *A Hybrid LSTM-Transformer Model for Anomaly Detection: Unraveling Hidden Patterns in Complex Data*, p. 352–359. New York, NY, USA: Association for Computing Machinery, 2025.
 - [11] J. A. Deja, S. Mayer, K. Čopić Pucihar, and M. Kljun, “A survey of augmented piano prototypes: Has augmentation improved learning experiences?,” *Proceedings of the ACM on Human-Computer Interaction*, vol. 6, p. 226–253, Nov. 2022.
 - [12] J. Liu, “Convolutional neural network-based human movement recognition algorithm in sports analysis,” *Frontiers in Psychology*, vol. 12, p. 663359, 2021.
 - [13] C. Lappe, M. Lappe, and P. E. Keller, “The influence of pitch feedback on learning of motor-timing and sequencing: A piano study with novices,” *PLOS ONE*, vol. 13, no. 11, p. e0207462, 2018.
 - [14] S. Furuya, M. Flanders, and J. F. Soechting, “Hand kinematics of piano playing,” *Journal of Neurophysiology*, vol. 106, no. 6, pp. 2849–2864, 2011.
 - [15] E. Nakamura, N. Ono, and S. Sagayama, “Merged-output hmm for piano fingering of both hands,” in *International Society for Music Information Retrieval Conference*, 2014.
 - [16] M. Hori, C. M. Wilk, and S. Sagayama, “Piano practice evaluation and visualization by hmm for arbitrary jumps and mistakes,” in *2019 53rd Annual Conference on Information Sciences and Systems (CISS)*, pp. 1–5, 2019.
 - [17] V. Krakovna and F. Doshi-Velez, “Increasing the interpretability of recurrent neural networks using hidden markov models,” 2016.
 - [18] L. Rimella and N. Whiteley, “Hidden markov neural networks,” *arXiv preprint arXiv:2004.06963*, 2025. Version 3, January 16, 2025.
 - [19] K. Abraham, E. Gassiat, and Z. Naulet, “Fundamental limits for learning hidden markov model parameters,” *IEEE Transactions on Information Theory*, vol. 69, p. 1777–1794, Mar. 2023.
 - [20] T. Xu, “Research on action recognition and feedback system in piano teaching based on deep learning,” in *2025 International Conference on Digital Analysis and Processing, Intelligent Computation (DAPIC)*, pp. 658–664, 2025.
 - [21] Y. Kim, J. Park, J. Bae, K. Kim, T. Kwon, A. Lerch, and J. Nam, “Pianovam: A multi-modal piano performance dataset,” in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025.
 - [22] Q. Gan, S. Wang, S. Wu, and J. Zhu, “Pianomotion10m: Dataset and benchmark for hand motion generation in piano performance,” 2025.
 - [23] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, and M. Grundmann, “Mediapipe hands: On-device real-time hand tracking,” 2020.
 - [24] R. W. Schafer, “What is a savitzky-golay filter? [lecture notes],” *IEEE Signal Processing Magazine*, vol. 28, pp. 111–117, July 2011.
 - [25] M. Schuster and K. Paliwal, “Bidirectional recurrent neural networks,” *IEEE Transactions on Signal Processing*, vol. 45, no. 11, pp. 2673–2681, 1997.

- [26] A. Graves and J. Schmidhuber, “Framewise phoneme classification with bidirectional lstm networks,” in *Proceedings. 2005 IEEE International Joint Conference on Neural Networks, 2005.*, vol. 4, pp. 2047–2052 vol. 4, 2005.
- [27] R. L. Smith, “Estimating tails of probability distributions,” *The Annals of Statistics*, vol. 15, no. 3, pp. 1174–1207, 1987.
- [28] T. Yoshida, “Asymptotic theory for extreme value generalized additive models,” 2025.
- [29] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, “Overcoming catastrophic forgetting in neural networks,” *Proceedings of the National Academy of Sciences of the United States of America*, vol. 114, no. 13, pp. 3521–3526, 2017.
- [30] L. E. Baum, T. Petrie, G. Soules, and N. Weiss, “A maximization technique occurring in the statistical analysis of probabilistic functions of markov chains,” *The Annals of Mathematical Statistics*, vol. 41, no. 1, pp. 164–171, 1970.
- [31] H. Do, J. Hwang, D. Han, S. J. Oh, and S. Yun, “What defines good reasoning in LLMs? dissecting reasoning steps with multi-aspect evaluation,” Oct. 2025. arXiv preprint arXiv:2510.20603.

Appendix

A Judge Scoring Rationales for the Base BiLSTM Two-Pass Augmented Model

The following cards summarize the broad-rubric LLM-as-a-judge rationales for the BiLSTM two-pass augmented model on all 20 human-annotated holdout videos. The score is shown on a 1–5 similarity scale.

Video 1

2/5

Weak overlap. The synthetic feedback correctly used hand-labeled positive language for the right hand and mentioned curvature and wrist level, but it contradicted the human reference by marking the left hand good despite reference concerns about rigid non-thumb fingers, insufficient curvature, and thumb/keybed position.

Video 2

1/5

Minimal overlap. The reference identified poor technique in both hands, including stiff fingers, flat index shape, shallow keybed position, inactive-finger tension, and strained wrist motion, while the synthetic feedback treated the clip as positive and therefore contradicted the main diagnostic intent.

Video 3

3/5

Moderate overlap. Both reference and synthetic feedback described generally good hand organization and curvature. The synthetic feedback missed the more specific reference refinements about deeper keybed position and residual inactive-finger tension.

Video 4

3/5

Moderate broad overlap. Both responses identified poor hand mechanics involving wrist and finger-shape problems. The synthetic feedback captured finger curvature and wrist-level instability, but missed the reference’s more specific lateral wrist rotation, shallow keybed position, pinky activation, thumb overstretch, and right-hand pinky collapse.

Video 5

2/5

Weak partial overlap. The synthetic feedback identified wrist and finger-shape instability, but it mislabeled the left hand as poor while the reference described left-hand wrist movement as fluid. It also missed the right-hand floating thumb, no keybed contact, flared inactive fingers, and insufficient vertical wrist movement.

Video 6

3/5

Moderate overlap. The synthetic feedback correctly treated the right hand as problematic and mentioned wrist instability and finger-shape issues, which broadly correspond to the reference’s right-hand wrist rigidity, collapse onto the thumb, flaring, and strain. It missed the thumb tuck and pinky-floating details, and it overemphasized left-hand inconsistency.

Video 7

4/5

Strong broad overlap. Both responses emphasized poor finger curvature and inefficient hand support. The synthetic left-hand flat-index diagnosis aligned with the reference’s outstretched index and middle fingers, and the right-hand low/unsupported hand shape broadly matched the reference’s inadequate curvature and rigid wrist mechanics, though thumb flaring and horizontal rotation were omitted.

Video 8

3/5

Moderate broad overlap. Both responses discussed poor hand organization, finger-shape problems, and wrist mechanics. The synthetic feedback captured unstable wrist/finger-curve families but missed the reference’s specific inactive-finger elevation, hand collapse onto the thumb, pinky compression, and excessive horizontal wrist rotation.

Video 9

4/5

Strong broad overlap. Both responses identified poor finger curvature and wrist-mechanics problems. The synthetic feedback’s flat finger and unstable wrist language broadly matched the reference’s poor curvature, pronounced wrist rotation, non-fluid motion, and wrist rigidity, although shallow keybed position, collapsed right-

hand fingers, and thumb strain were not fully captured.

Video 10

3/5

Moderate overlap. Both responses identified finger-curve and wrist-mechanics concerns, especially in the left hand. The synthetic feedback missed the reference's more specific pinky flaring, active middle finger tension, and the right hand's mostly adequate curvature with only mild strain/rotation.

Video 11

2/5

Weak overlap. The synthetic feedback captured broad wrist/finger-shape instability, but the reference emphasized left-hand inactive-finger flaring with otherwise fluid wrist motion and right-hand ring/pinky strain with wrist rigidity. The synthetic feedback did not clearly recover those hand-specific distinctions.

Video 12

4/5

Strong broad overlap. Both responses described poor finger shape and wrist-mechanics issues. The synthetic feedback's inconsistent finger curve and unstable wrist level broadly matched the reference's collapsed over-curved index finger, shallow keybed position, collapsed wrist, inactive middle/ring fingers, and rigid right-hand trajectories, though the exact mechanisms were less specific.

Video 13

2/5

Weak overlap. The synthetic feedback mentioned general finger curve and wrist alignment issues, but the reference distinguished a mostly good left hand with strained ring/pinky fingers from a poor right hand with flat index-through-ring shape, visible strain, and excessive horizontal wrist rotation.

Video 14

2/5

Weak overlap. The reference described mostly good overall hand state with minimal movement but noted shallow keybed and wrist rigidity. The synthetic feedback gave broad wrist/finger observations without recovering the low-motion con-

text or shallow keybed diagnosis clearly.

Video 15

1/5

No meaningful overlap. The reference identified severe issues in both hands, including flat rigid fingers, poor hand curvature, wrist rigidity, collapse, flaring, and low thumb/keybed position. The synthetic feedback did not capture these core mechanisms adequately.

Video 16

2/5

Weak overlap. The synthetic feedback captured broad finger-shape and wrist/trajectory concepts, but the reference was more specific: left pinky flat/strained, rigid index-ring trajectories, relaxed wrist, and right ring-finger activation despite adequate hand/wrist curvature.

Video 17

3/5

Moderate overlap. Both responses shared broad concerns about hand shape, wrist mechanics, and movement efficiency. The synthetic feedback aligned partly with insufficient curvature and vertical wrist movement but missed the left hand lifting off the piano and the right thumb tucking/flaring with strained fingers.

Video 18

3/5

Moderate overlap. Both responses captured broad finger-shape, wrist-mechanics, and thumb-position families. The synthetic feedback did not fully recover shallow keybed position, rigid middle/index trajectories, tucked left thumb, or excessive right horizontal wrist rotation.

Video 19

3/5

Moderate overlap. The synthetic feedback captured broad wrist and finger-trajectory issues, while the reference emphasized clumped fingers, insufficient individual activation, flared thumb, flat active middle finger, shallow thumb keybed, and inward wrist tilt. The overlap was conceptual but not mechanism-specific.

Video 20

2/5

Weak-to-moderate overlap. The synthetic feedback captured broad wrist/finger-shape concerns, but the reference's main issues were left-hand inactive pinky/ring tension, extended strained middle finger, excessive horizontal wrist rotation, and right-hand thumb tucking with insufficient vertical wrist tilt. Several reference-specific details were missed.

B Raw Synthetic Feedback

This section reports final synthetic feedback examples from the base BiLSTM + two-pass + augmented model on the external holdout videos. Each card includes the judge similarity score, selected feedback branch, and generated feedback text.

Video 1

2/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. The hand shape stays compact and the fingertip path is controlled, helping the motion remain relaxed and efficient. Right hand (overall good): The wrist level stays steady and the wrist rotation looks controlled, which supports efficient movement through the phrase.

Video 2

1/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. The hand shape stays compact and the fingertip path is controlled, helping the motion remain relaxed and efficient.

Video 3

3/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. Right hand (overall good): The hand shape stays compact and the fingertip path is controlled, helping the motion remain relaxed and efficient.

Video 4

3/5

anomalous

Left hand (overall poor): The middle finger looks too flat, which can reduce control. Keep a

rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The index finger looks too flat, which can reduce control. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 5

2/5

anomalous

Left hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. The wrist alignment changes too abruptly, which can make turns feel larger and less prepared. Prepare the wrist motion earlier and keep the turn smaller.

Video 6

3/5

anomalous

Left hand (overall inconsistent): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The ring finger looks too flat, which can reduce control and make the hand feel unsupported. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 7

4/5

anomalous

Left hand (overall inconsistent): The index finger looks too flat, which can reduce control. Keep a rounder finger shape as the finger approaches the key. The wrist level is also unstable, which

can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The hand sits too low, which can make the hand collapse. Bring the hand slightly more above the keys and avoid sinking into the keyboard. The finger curve is inconsistent, which makes the hand shape less reliable. Aim for a more stable rounded shape through each finger.

Video 8

3/5

anomalous

Left hand (overall poor): The middle finger looks too flat, which can reduce control. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control.

Video 9

4/5

anomalous

Left hand (overall poor): The middle finger looks too flat, which can reduce control and make the hand feel unsupported. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 10

3/5

anomalous

Left hand (overall inconsistent): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is also unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier

through the movement. The wrist alignment changes too abruptly, which can make turns feel larger and less prepared. Prepare the wrist motion earlier and keep the turn smaller.

Video 11

2/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. Right hand (overall good): The wrist level stays steady and the wrist rotation looks controlled, which helps the hand move efficiently.

Video 12

4/5

anomalous

Left hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The index finger looks too flat, which can reduce control and make the hand feel unsupported. Keep a rounder finger shape as the finger approaches the key. The wrist alignment changes too abruptly, which can make turns feel larger and less prepared. Prepare the wrist motion earlier and keep the turn smaller.

Video 13

2/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. Right hand (overall good): The wrist level stays steady and the wrist rotation looks controlled, which helps the hand move efficiently.

Video 14

2/5

anomalous

Left hand (overall poor): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays

rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 15

1/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. Right hand (overall good): The wrist level stays steady and the wrist rotation looks controlled, which helps the hand move efficiently.

Video 16

2/5

non anomalous

Left hand (overall good): The wrist level stays steady and the finger curve remains organized, which supports control without extra strain. Right hand (overall good): The wrist level stays steady and the wrist rotation looks controlled, which helps the hand move efficiently.

Video 17

3/5

anomalous

Left hand (overall inconsistent): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist sits too high, which can make the fingers work harder than necessary. Let the hand settle to a more natural level above the keys.

Video 18

3/5

anomalous

Left hand (overall poor): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The middle finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays

rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 19

3/5

anomalous

Left hand (overall poor): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The index finger looks too flat, which can reduce control and make the hand feel unsupported. Keep a rounder finger shape as the finger approaches the key. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

Video 20

2/5

anomalous

Left hand (overall poor): The thumb finger shape looks over-curved or locked, which can create excess tension. Release the finger joints slightly so the shape stays rounded without gripping. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement. Right hand (overall poor): The finger curve is inconsistent, which makes the hand shape less reliable from note to note. Aim for a more stable rounded shape through each finger. The wrist level is unstable, which can interrupt smooth control. Keep the wrist level steadier through the movement.

C Raw Human-Annotated Reference Feedback

The following cards contain the raw human-annotated reference feedback for each of the 20 external holdout videos. Annotations were produced by the author in the capacity of an advanced pianist prior to any model comparison, under a predefined anti-bias protocol. All videos were evaluated with audio muted. These annotations served as the reference against which synthetic pipeline feedback was compared in the LLM-as-a-judge evaluation.

Video 1

Left hand (overall poor): Non-thumb finger joints are rigid, with insufficient index and mid-

dle finger curvature producing stiff movement. Wrist height is stable. The thumb requires counterclockwise rotation to achieve deeper keybed contact.

Right hand (overall good): Slight curvature is present across non-thumb joints, which is more favorable than the left hand. The thumb and pinky remain relatively relaxed, supporting more fluid movement. The right thumb should be positioned deeper in the keybed for greater control.

Video 2

Left hand (overall poor): Finger joints are stiff, producing linear, rigid movement. The index finger remains notably flat; increased curvature is needed. The hand sits too shallow in the keybed.

Right hand (overall poor): The ring and pinky fingers tighten and straighten even when not playing. The hand should be positioned deeper in the keybed. Wrist movement appears strained, and the index finger requires more curvature.

Video 3

Left hand (overall good): Hand posture and curvature are well-maintained. The hand could be positioned deeper in the keybed. Residual tension is visible in the middle through pinky fingers when they should remain relaxed.

Right hand (overall good): Hand posture, curvature, and non-playing finger relaxation are all well-maintained. The hand could be positioned deeper in the keybed.

Video 4

Left hand (overall poor): The wrist is rotated laterally to the left and is not aligned with the forearm. The pinky remains tight and activated when not playing. Fingers are over-curved and sit too shallow in the keybed. The thumb is overstretched, and the hand appears to be shaped around thumb extensions, creating visible strain.

Right hand (overall poor): The pinky is collapsed, with adjacent fingers reaching over it. The wrist exhibits excessive horizontal-plane rotation and is not aligned with the forearm. The hand sits too shallow in the keybed.

Video 5

Left hand (overall good): Wrist movement is fluid with no visible strain. Finger movement is inefficient due to excessive horizontal forward-backward motion. The pinky and ring finger are not consistently relaxed.

Right hand (overall poor): The thumb is left floating and tense with no keybed contact. The pinky and ring fingers are flared and tense even when not playing. The wrist is too rigid with insufficient vertical movement, placing excess demand on the fingers.

Video 6

Left hand (overall good): The thumb is left floating and tense at points. The hand is well-positioned in the keybed. Finger curvature is adequate.

Right hand (overall poor): Wrist rigidity causes the hand to collapse onto the thumb. Fingers are flared and appear strained. The thumb tucks awkwardly under the index finger. The pinky is occasionally left floating and flared.

Video 7

Left hand (overall poor): The index and middle fingers are left outstretched with poor curvature. The wrist is obscured in the video but likely rigid, contributing to poor motion efficiency. The thumb is rigid, insufficiently curved, and sits too shallow in the keybed.

Right hand (overall poor): The hand lacks adequate curvature. The wrist is rigid and exhibits excessive horizontal rotation to the right, with the wrist not aligned with the forearm. The thumb is turned upward and flares outward.

Video 8

Left hand (overall poor): The middle and index fingers remain tight, flared, and elevated even when not playing. The hand collapses onto the thumb due to rigid wrist posture. The fingers are curved, but the center of the hand is not aligned with the overall finger curvature.

Right hand (overall poor): The middle through pinky fingers show visible strain when they should be relaxed. The pinky is not aligned with the wrist curvature, with the joint attaching the pinky to the hand appearing compressed. The wrist is not aligned with the forearm due to

excessive horizontal rotation.

Video 9

Left hand (overall poor): Finger curvature is poor across all fingers. Pronounced wrist rotation produces non-fluid movement. Wrist rigidity contributes to inefficient motion overall.

Right hand (overall poor): The middle through pinky fingers are collapsed. The hand sits too shallow in the keybed with little curvature. The thumb is strained and sits too shallow on the keys.

Video 10

Left hand (overall poor): The pinky is strained and flared. Wrist movement is rigid. The middle finger is active and tense when it should be relaxed.

Right hand (overall good): The middle finger is active and strained when it should be relaxed. Finger curvature across all fingers is adequate. The wrist is slightly strained and rotated.

Video 11

Left hand (overall good): The ring, middle, and index fingers are tight and flared when they should be relaxed. Wrist movement is fluid and well-controlled. The hand sits too shallow in the keybed.

Right hand (overall poor): The ring finger is tight and flared when it should be relaxed. Wrist rigidity contributes to inefficient hand motion. The pinky shows excessive strain.

Video 12

Left hand (overall poor): The index finger is collapsed through over-curvature. The hand sits too shallow in the keybed. The wrist is collapsed and lacks activation. The middle and ring fingers are activated when they should be relaxed.

Right hand (overall poor): Finger curvature is more adequate. Finger trajectories are rigid, with limited fluid or active motion in the ring and pinky fingers.

Video 13

Left hand (overall good): Hand and finger curvature are adequate. The ring and pinky fingers are strained and over-extended. The wrist

is slightly rigid.

Right hand (overall poor): The index through ring fingers are notably flat with poor overall hand curvature. The middle and index fingers show visible strain. The wrist is rigid with excessive horizontal-axis rotation.

Video 14

Left hand (overall good): Non-playing fingers remain relaxed. The wrist is rigid and the hand sits too shallow in the keybed. Overall movement in the clip is minimal.

Right hand (overall good): The pinky is slightly activated when it should be relaxed. The wrist is rigid and the hand sits too shallow in the keybed. Overall movement in the clip is minimal.

Video 15

Left hand (overall poor): The pinky is outstretched and strained. The ring through index fingers are flat and rigid with poor hand and finger curvature. Wrist rigidity significantly impairs the movement trajectories of the ring and pinky fingers.

Right hand (overall poor): The pinky is collapsing. The fingers are strained and flat, spreading and tensing even when they should be relaxed. The thumb is flared and positioned too low on the keybed.

Video 16

Left hand (overall good): The pinky is flat and strained. The index, middle, and ring finger trajectories are rigid, contributing to inefficient movement. The wrist is adequately relaxed.

Right hand (overall poor): Hand and wrist curvature are adequate. Ring finger movement is rigid and slightly activated when it should be relaxed.

Video 17

Left hand (overall good): Finger trajectories are fluid. Insufficient hand curvature and limited vertical wrist movement contribute to inefficient motion. The hand is lifted off the piano midway through the clip to rest.

Right hand (overall poor): The thumb trajectory is poor due to tucking and outward flaring. The index through pinky fingers are strained. Increased vertical wrist tilt is needed.

Video 18

Left hand (overall poor): Finger curvature is adequate. The hand sits too shallow in the keybed. The middle and index finger trajectories are rigid. The thumb is tucked in and insufficiently active.

Right hand (overall poor): The thumb is strained and flared. Excessive horizontal wrist rotation is present. Hand and finger curvature are adequate.

Video 19

Left hand (overall poor): Vertical wrist tilt motion is well-executed. The fingers are clumped together with insufficient individual activation, producing poor finger trajectories and inefficient motion. The thumb is flared and slightly active when it should be relaxed.

Right hand (overall poor): The middle finger is flat, active, and rigid when it should be relaxed. The thumb sits too shallow in the keybed; increased inward wrist tilt would improve keybed depth.

Video 20

Left hand (overall poor): The pinky and ring fingers are tensed and active when they should be relaxed. The middle finger is extended and strained. Wrist movement is rigid and includes excessive horizontal-plane rotation.

Right hand (overall good): The thumb is tucked too far under. Vertical wrist tilt motion is insufficient. Finger trajectories are fluid with good hand and finger curvature.